New York City Airbnb Data Analysis

From Raw Data to Actionable Business Insights

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Project Architecture & Workflow

How raw data was transformed into actionable insights.

1. Raw Data

Kaggle NYC Airbnb Open Data (2019). Over 100,000+ rows of listings.

01

2. Python Data Preparation

Pandas used for EDA, handling missing values, outlier removal, and date formatting

02

3. Power BI Analysis

DAX measures created for dynamic KPIs. Interactive dashboards built for reporting.

03

Project Overview

The Concept: Analyzing Airbnb platform data to understand

- Listing distribution
- Pricing patterns
- Customer engagement through reviews
- Availability
- Booking behavior
- House rules & cancellation policies

Main Question?

How can Airbnb listing data be transformed into insights that support better pricing, listing management, and customer experience decisions?

Business Objectives

1. Market Structure

Understand how Airbnb supply is distributed by:

- Neighborhood Group
- Room Type

2. Pricing

Evaluate pricing patterns using :

- Average Price
- Service Fee
- Minimum Nights
- Price differences across locations and room types

Business Objectives

3. Customer Engagement

Understand listing engagement through:

- Listings with Reviews
- Review Rate
- Number of Reviews
- Availability
- Instant Booking

4. Policies & Operations

Evaluate operational patterns through:

- Cancellation Policy
- House Rules
- Minimum Nights

Our Goal

Identify patterns, problems and opportunities that can support better business decisions.

Business Problems

Problems We Aim to Investigate:

Problem 1: Uneven Market Distribution

Are Airbnb listings heavily concentrated in specific areas?

Problem 2: Pricing Uncertainty

Does location create a meaningful difference in listing prices?

Problem 3: Availability & Booking

Are some room types more available than others?

Problem 4: Operational Policies

Are some neighborhoods or room types associated with different minimum stays or restrictive policies?

Dataset Overview

The dataset contains listing-level information covering:

- Listing information
- Location
- Reviews
- Booking preferences
- House rules
- Host information
- Pricing
- Availability
- Cancellation policies

Key Fields

- id
- neighbourhood_group
- price
- minimum_nights
- last_review
- instant_bookable
- Room Type
- host_id
- neighbourhood
- service_fee
- number_of_reviews
- availability_365
- cancellation_policy

Data Cleaning & Quality Assessment

Cleaning Principles

Each cleaning decision was evaluated based on:

Data Quality + Analytical Importance + Business Relevance

1. Missing Values

- Highly incomplete columns were removed when they had limited analytical value.
- Categorical missing values were handled using Unknown where appropriate.
- Review-related missing values were treated carefully to avoid creating misleading information.

2. Duplicates

- Listing IDs were checked to ensure the uniqueness of listings.

Data Cleaning

3. Data Types

The dataset was checked to ensure appropriate:

- Numeric data types
- Date formats
- Boolean values
- Categorical values

4. Outliers

Potential outliers were reviewed in:

- Price
- Service Fee
- Minimum Nights
- Number of Reviews

The goal: was to differentiate between data errors and real extreme listings.

Python Data Preparation Pipeline

Raw Data → ***Data Inspection*** → ***Missing Value Analysis***

Data Cleaning → ***Data Type Correction*** → ***Feature Preparation***

Clean Dataset → ***Power BI***

Python was used as the data preparation layer, while **Power BI** was used for modeling, DAX calculations, visualization, and business analysis.

Power BI Data Model

Data Model Components:

The Power BI model contains:

airbnb_cleaned
Fact Table

Calendar
Time Analysis

house_rules_table
House rule analysis

Measures_table
Centralized DAX measures

Why Use a Data Model?

The model allows us to:

- Centralize calculations
- Create reusable DAX measures
- Apply filters consistently
- Separate data, calculations, and presentation
- Support interactive analysis

DAX & KPIs

Key Measures

Total Listings =DISTINCTCOUNT(airbnb_cleaned[id])

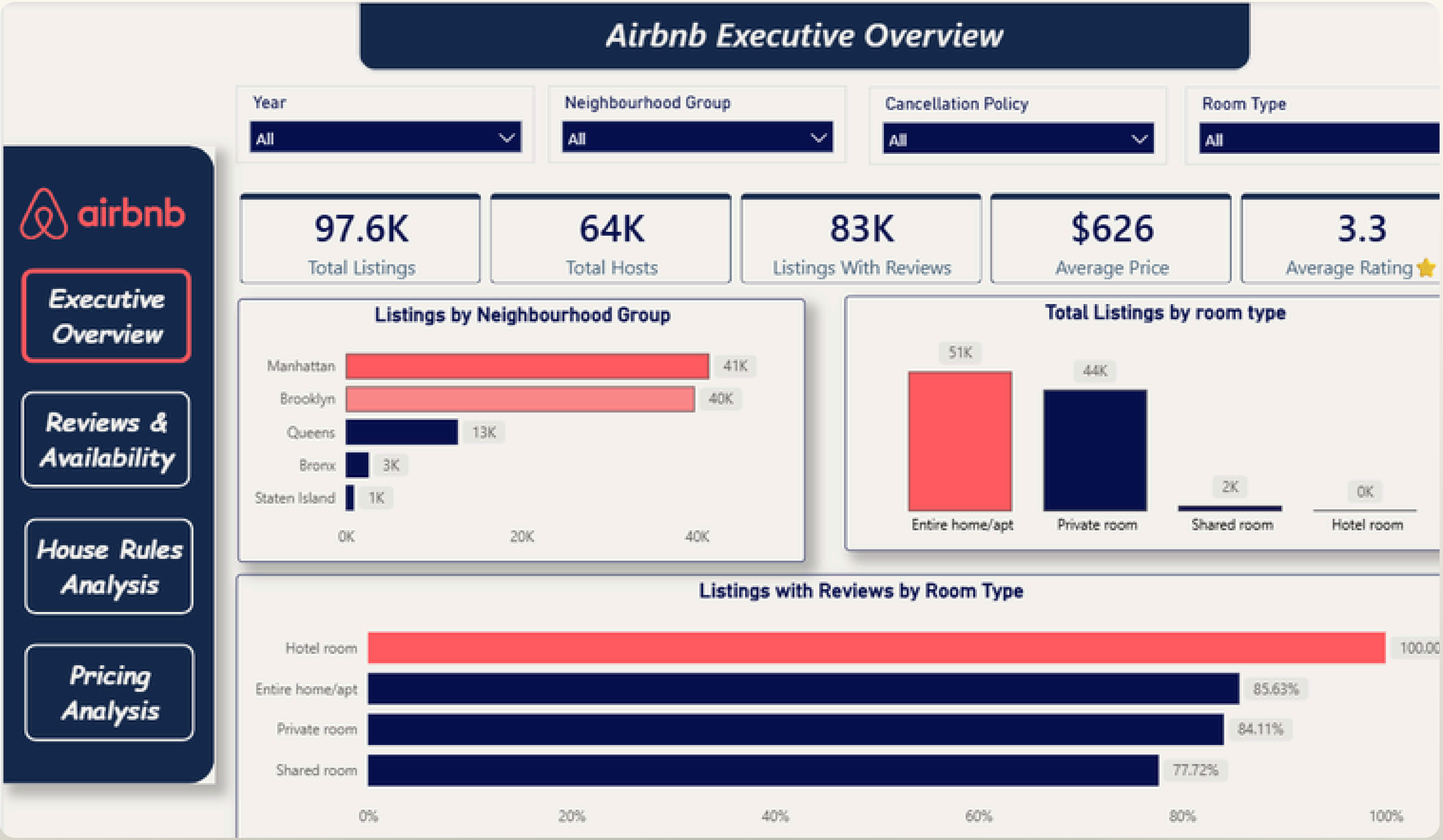
Average Price = AVERAGE(airbnb_cleaned[price])

Listings With Reviews = CALCULATE([Total Listings], airbnb_cleaned[review_status] = "Has Reviews")

Average Minimum Nights = AVERAGE(airbnb_cleaned[minimum nights])

host id2 = CONCATENATE(CONCATENATE(airbnb_cleaned[host name],airbnb_cleaned[long]),airbnb_cleaned[lat])

Executive Overview



Executive Overview

KPI Cards

- Total Listings
- Average Rating
- Listings With Reviews
- Average Price
- Total Hosts

Visual 1: Listings by Neighborhood

Business Insight: Listings are highly concentrated in Manhattan and Brooklyn, indicating that supply is strongly concentrated in a limited number of markets.

Business Implication: These markets represent the largest supply concentration and should receive greater attention when evaluating competition and market strategy.

Visual 2: Listings by Room Type

Business Insight: The market is dominated by Entire Homes and Private Rooms, while Shared Rooms and Hotel Rooms represent a very small portion of the listing portfolio.

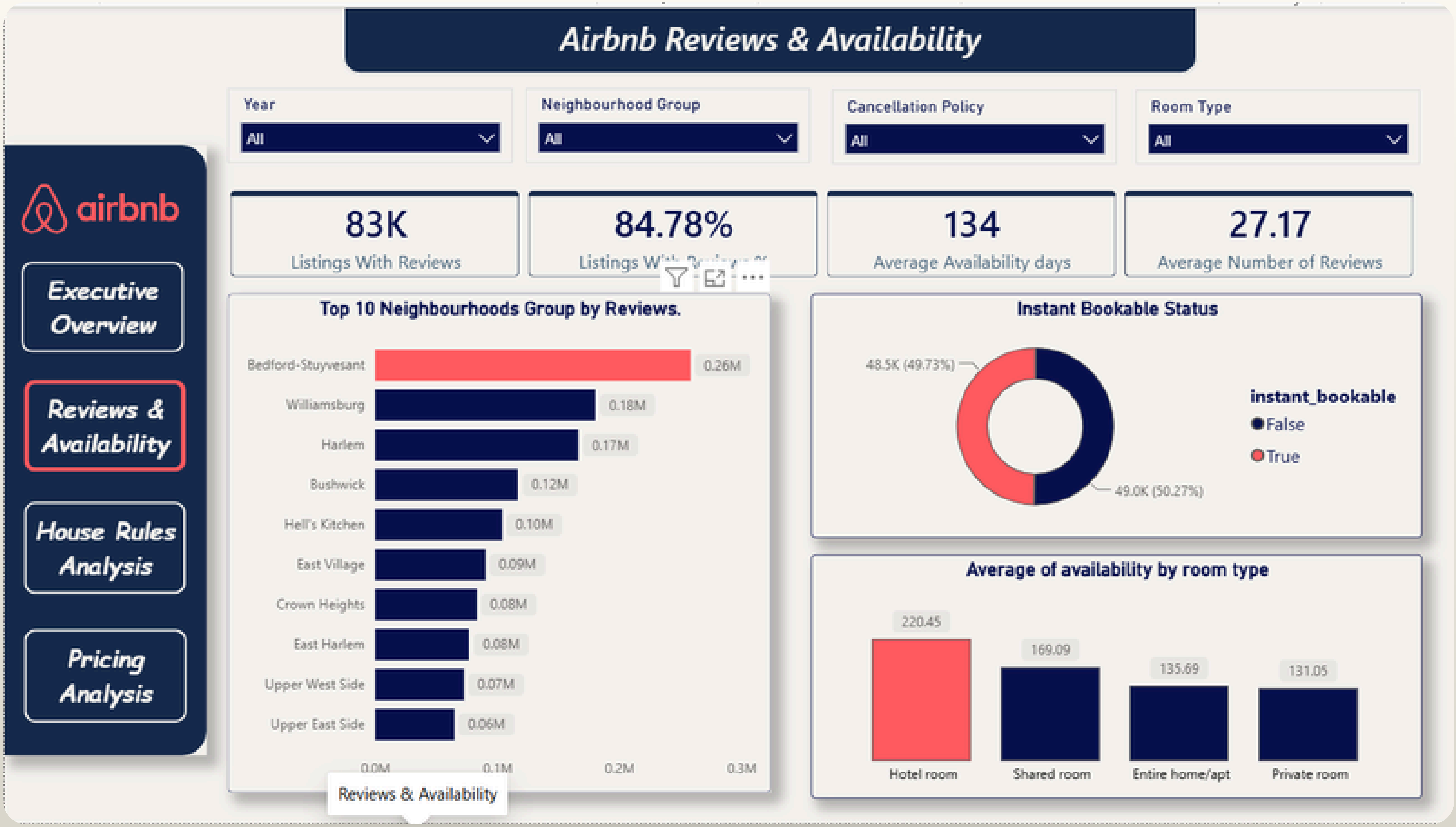
Business Implication: The platform's supply is primarily built around residential accommodation rather than hotel-style inventory

Visual 3: Listings With Reviews by Room Type

Business Insight: Hotel Rooms listings have the highest review coverage at 100%, while Shared Room have the lowest at 77%.

Business Implication: Review engagement varies considerably by room type, with Shared Room listings showing a larger gap in customer feedback compared with other Room types.

Reviews & Availability



Reviews & Availability

KPI Cards

- Listings With Reviews
- Average Availability
- Listings With Reviews %
- Average Number of Reviews

Visual 1: Listings With Reviews

Business Insight: Approximately 85% of listings have reviews, indicating strong customer engagement across the marketplace.

Business Implication: Reviews represent an important indicator of customer interaction and listing activity.

Visual 2: Availability by Room Type

Business Insight: Hotel and Shared Room listings show substantially higher availability than Private Rooms and Entire Homes.

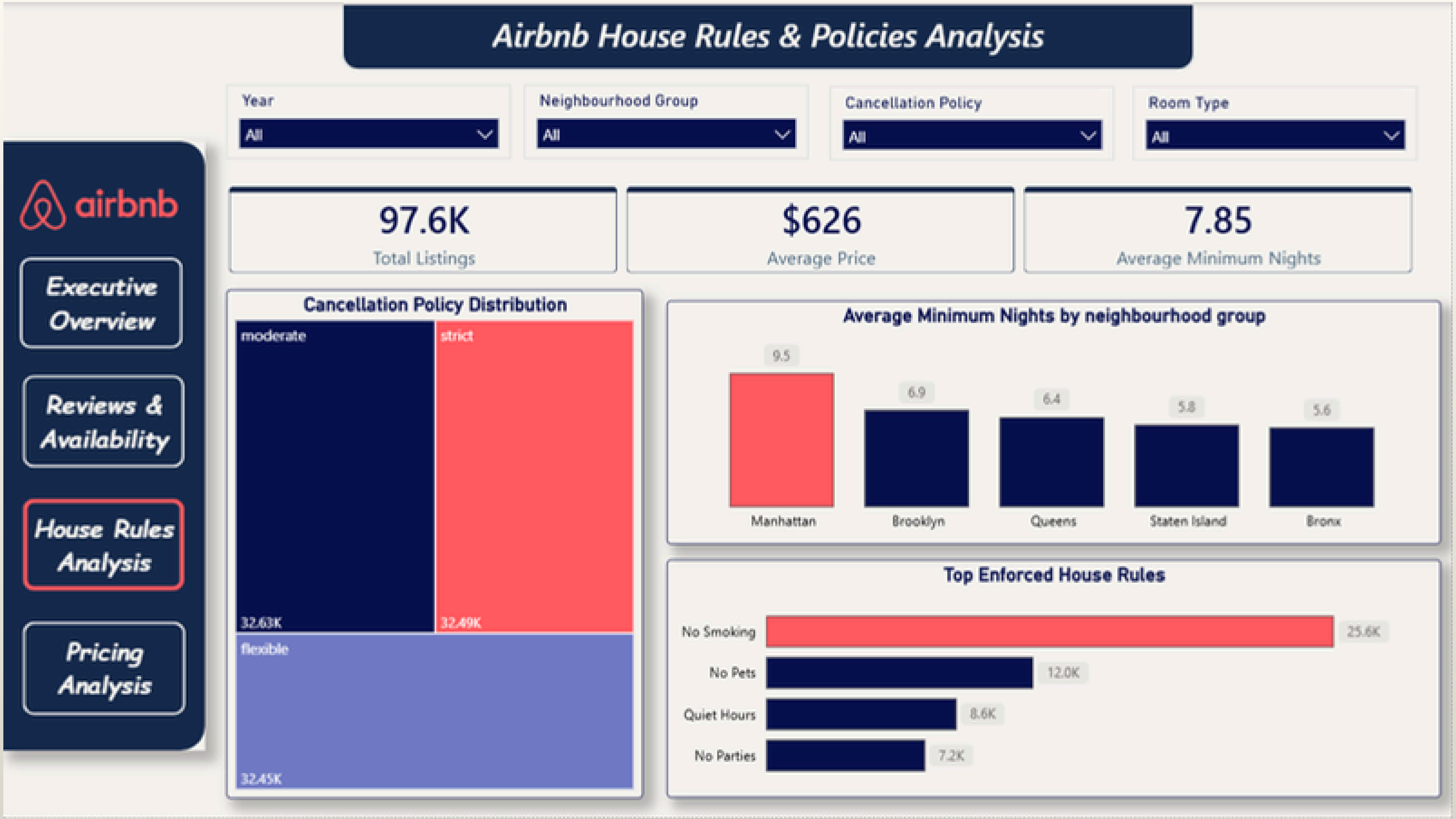
Business Implication: Higher availability may indicate differences in demand, booking frequency, or inventory utilization.

Visual 3: Instant Booking

Business Insight: Instant Booking is almost evenly split across listings, indicating that hosts are divided between immediate booking and host-confirmation models..

Business Implication: Further analysis could determine whether Instant Booking is associated with stronger customer engagement or listing performance.

House Rules & Policies



House Rules & Policies

KPI Cards

- Total Listings
- Average Price
- Average Minimum nights

Visual 1: Cancellation Policy

Business Insight: Flexible, Moderate, and Strict cancellation policies are almost evenly distributed across the marketplace.

Business Implication: There is no single dominant cancellation strategy, suggesting that hosts use different approaches depending on their business model.

Visual 2: Minimum Nights by Neighborhood

Business Insight: Manhattan has the highest average minimum-stay requirement, while the other neighborhoods have noticeably lower requirements.

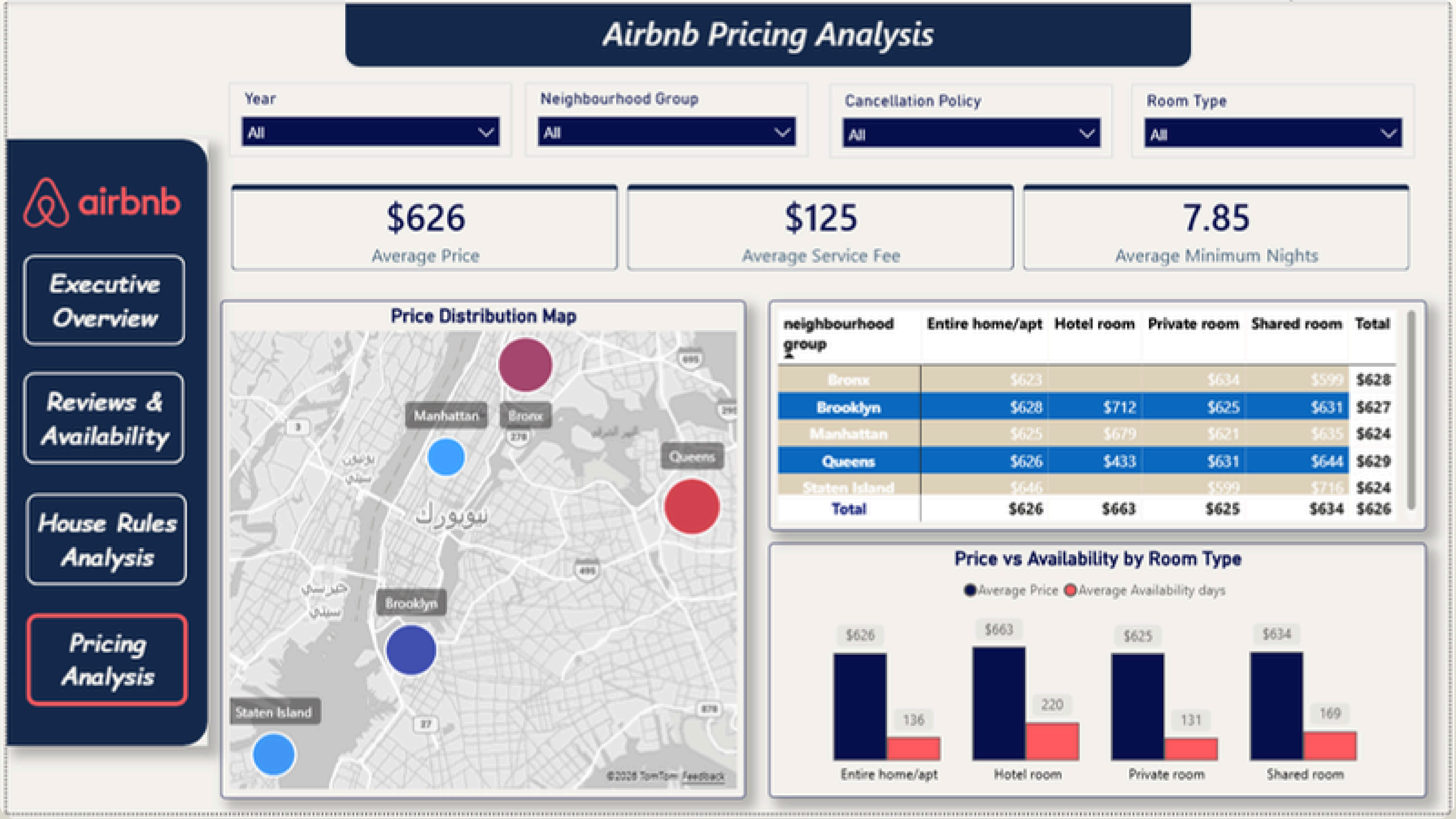
Business Implication: Higher minimum stays may reduce booking flexibility and should be evaluated against availability and customer engagement..

Visual 3: Top Enforced House Rules

Business Insight: No Smoking is by far the most common enforced house rule, followed by No Pets, Quiet Hours, and No Parties

Business Implication: These rules represent the most common operational restrictions and should be clearly communicated to guests to reduce potential conflicts.

Pricing Analysis



Pricing Analysis

KPI Cards

- Average Service Fee
- Average Price
- Average Minimum nights

Visual 1: Price Map

Business Insight: Average listing prices vary only slightly across neighborhood groups, ranging approximately from $624 to $629.

Business Implication: Neighborhood alone does not appear to be a strong differentiator of average price in this dataset.
Therefore, pricing decisions should consider additional factors such as:
Room Type, Availability, Reviews, Property characteristics, Demand.

Visual 2: Pricing by Neighbourhood & Room Type

Business Insight: Overall neighbourhood prices are relatively stable, but significant price differences appear when comparing specific room types across neighbourhoods

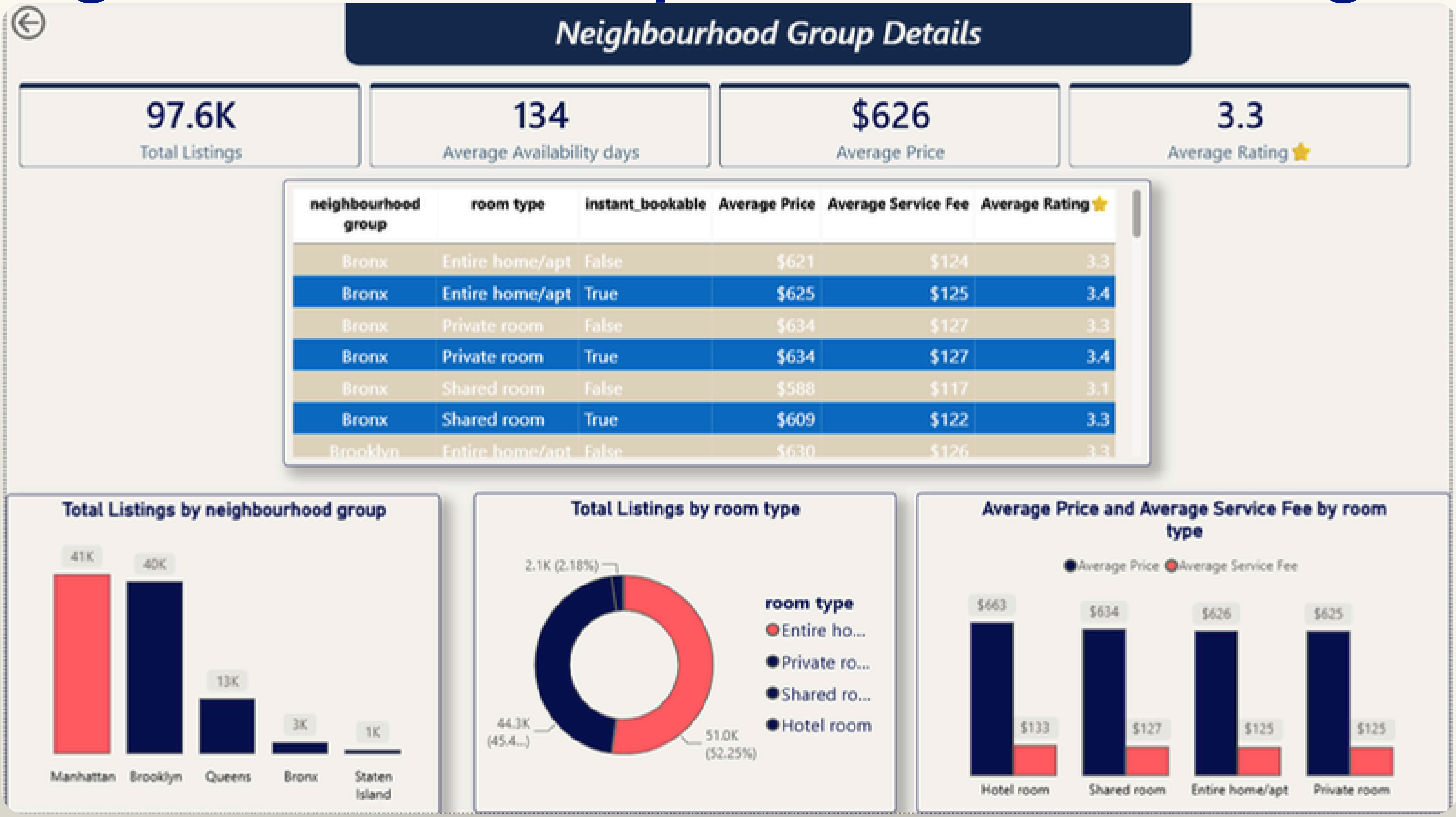
Business Implication: Pricing strategies should be tailored to both neighbourhood and room type rather than relying on neighbourhood-level averages alone.

Visual 3: Price vs Availability by Room Type

Business Insight: Hotel rooms have the highest average price at $663 and the highest average availability at 220 days, while private rooms have the lowest availability at 131 days.

Business Implication: The hotel segment should be investigated further to understand whether its high availability reflects excess inventory, lower demand, or a different operating model before making pricing decisions.

Neighbourhood Group Details(Drill-theough)



Drill-through Page

KPI Cards

- Total Listings
- Average Availability
- Average Price
- Average Rating

Visual 1: Total Listings by Neighbourhood Group

Business Insight: Listing inventory is heavily concentrated in Manhattan and Brooklyn, with approximately 41K and 40K listings respectively, while Queens has around 13K and the remaining neighbourhoods have substantially fewer listings.

Business Implication: Manhattan and Brooklyn represent the largest markets by listing volume, making them the primary areas for inventory and market-performance monitoring.

Visual 2: Total Listings by Room Type

Business Insight: Entire homes/apt and private rooms dominate the listing inventory, accounting for approximately 52% and 45% of listings respectively, while shared and hotel rooms represent only a small portion of the inventory.

Business Implication: Business strategies should primarily focus on the dominant entire-home and private-room segments, while smaller segments may require more targeted strategies rather than broad market-wide decisions.

Visual 3: Average Price & Average Service Fee by Room Type

Business Insight: Hotel rooms have the highest average price ($663) and service fee ($133), while private rooms have the lowest average price ($625) and service fee ($125).

Business Implication: The hotel segment generates higher revenue per listing but also carries higher customer-facing fees, making it a segment worth monitoring for pricing competitiveness and customer acceptance

Key Business Findings

1. Supply Concentration

Manhattan and Brooklyn dominate listing supply.
→ These markets represent the highest concentration of supply and competition.

2. Pricing Is Not Driven by Neighbourhood Alone

Average prices across neighborhood groups are very similar.
→ Pricing strategy should consider additional listing characteristics.

3. Strong Review Coverage

Approximately 85% of listings have reviews.
→ Customer feedback is an important marketplace engagement indicator.

Key Business Findings

4. Availability Differs by Room Type

Hotel and Shared Room listings have significantly higher availability.
→ These segments may require further demand and pricing analysis.

5. Minimum Stay Differences

Manhattan has a substantially higher average minimum stay.
→ Minimum-stay strategy varies considerably by location.

6. No Dominant Cancellation Policy

Flexible, Moderate, and Strict policies have similar distributions.
→ There is no universal cancellation strategy across the marketplace.

Recommended Actions

1. Focus on High-Supply Markets

Monitor competition and inventory dynamics in Manhattan and Brooklyn.

2. Move Beyond Location-Based Pricing

Evaluate room type, availability, reviews, and property characteristics when developing pricing strategies.

3.Investigate High-Availability Segments

Further investigate Hotel and Shared Rooms to determine whether high availability reflects excess inventory, demand patterns, or operating differences.

Recommended Actions

4. Improve Review Engagement

Investigate the lower review coverage among Hotel Room listings and identify opportunities to increase guest feedback.

5. Evaluate Minimum-Stay Policies

Assess whether higher minimum-stay requirements affect availability and customer engagement, particularly in Manhattan.

Limitations

Analytical Limitations

- The **year** variable is based on **Last Review**, not listing creation date.
- The dataset does not directly provide actual revenue.
- Occupancy data is not directly available.
- Listing price does not necessarily represent the final transaction price.
- Availability does not directly measure demand.
- Observed relationships do not necessarily imply causation.

Why This Matters?

The analysis identifies patterns and opportunities, but additional business data would be required to confirm causal relationships and make final strategic decisions.

Conclusion

The analysis shows that Airbnb's marketplace is characterized by:

- Strong listing concentration in Manhattan and Brooklyn.
- A market dominated by Entire Homes and Private Rooms.
- High review coverage across listings.
- Limited neighborhood-level differences in average price.
- Significant availability differences between room types.
- Different minimum-stay strategies across neighborhoods.
- Diverse cancellation policies.
- Strong emphasis on guest behavior restrictions.

Final Business Perspective

The analysis moves beyond describing Airbnb listings to identifying where supply is concentrated, where market behavior differs, and where further investigation can support better pricing, inventory, and operational decisions.

Thank You

Questions & Discussion